2024 Spring 生化学 2 English, Home work#5(250421) dead line: 23:59, April 28, 2025

学号:

姓名:

Q1. 判断题（请在括号里画√或 X, 1 分/题, 共 35 分）

(1) Hexokinase use ATP to convert Glucose to Glucose 6-hosphate. ( )
(2) Phosphofructokinase-1 (PFK-1) use GTP to convert Fructose 6-phosphate to Fructose 1,6-bisphosphate. ( )
(3) Aldolase makes 2 molecules of Glyceraldehyde 3-phospate from Fructose 1,6-bisphosphate. ( )
(4) The amount of free NADH is much lower than free NAD+ in cytoplasm. ( )
(5) Enolase converts 2-Phosphoglycerate to Phosphoenolpyruvate (PEP). ( )
(6) Enolase use Mn^{2+} as a cofactor. ( )
(7) Pyruvate kinase uses PEP and ADP to make pyruvate and ATP. ( )
(8) There are 4 irreversible steps in Glycolysis. ( )
(9) Tumor is aerobic condition. Therefore, they use glycolysis a lot. ( )
(10) In tumor, expression of some enzymes such as hexokinase is upregulated. ( )
(11) In the PET imaging, glucose analog containing 19F is used, and catalyzed by several enzymes. ( )
(12) Phosphorolysis provides Glucose from Glycogen. ( )
(13) UDP-glucose is the intermediate of Galactose to Glucose 1-phospate. ( )
(14) Fructose enters the Gycolytic pathway in the similar way in both Kidney and Liver. ( )
(15) In anaerobic condition, NAD+ can be provided by converting Pyruvate to Lactate, lowering the pH of the cell. ( )
(16) In fermentation, there are no net consumption of NAD/NADH. ( )
(17) Conversion of Pyruvate to Ethanol produces CO2 as byproduct. ( )
(18) Gluconeogenesis is just the reverse of Glycolysis. ( )
(19) ΔG of the 10 steps of Glycolysis are all near zero. ( )
(20) Pyruvate carboxylase converts Pyruvate to Oxaloacetate using ATP and biotin. ( )
(21) PEP carboxykinase uses ATP to convert Oxaloacetate to Phosphoenolpyruvate (PEP). ( )

(22) Oxaloacetate (OAA) cannot pass the mitochondria membrane. Therefore, OAA is converted to Malate in Liver. ( )

(23) NADH concentration is high in Mitochondria and low in cytoplasm, facilitating the conversion between Oxaloacetate and Malate. ( )

(24) Fructose 1,6-bisphosphatase (FBPase-1) consumes H_2O to convert Fructose 1,6-bisphosphate to Fructose-6-phosphate. ( )

(25) Hexokinase consume H2O to convert Glucose 6-phosphate to Glucose. ( )

(26) With one cycle of Glycolysis and Gluconeogenesis, one can obtain 2ATP and 2GTP. ( )

(27) In the Cori cycle, Glucose is converted to Lactate in Muscle. Then, Lactate is converted back to Glucose in Liver. Therefore, Glucose consumption and Gluconeogenesis occur in different organs. ( )

(28) Pentose phosphate pathway provides Ribose 5-phosphate and NADPH. ( )

(29) Ribose 5-phosphate is the source of RNA. ( )

(30) NADPH contributes to protect cells against ROS (Reactive Oxygen Species). ( )

(31) GSSG/GSH is the counter partner of NADPH in ROS protection. ( )

(32) The “non-oxidativee phase” of Pentose phosphate pathway produces hexose phosphate from pentose phosphate. ( )

(33) Some steps of the “non-oxidativee phase" of Pentose phosphate pathway are irreversible. ( )

(34) Transketolase use TPP (Tiamine pyrophospate) as cofactor. ( )

(35) Transaldolase also use TPP as cofactor. ( )

Q2. Write down enzymes and substrate/product of all Glycolysis 10 steps (Please use arrow to connect each steps) (各 1 分、共 10 分)

Q3. Write down enzymes and substrate/product of 1st bypass process of Gluconeogenesis (from pyruvate to PEP) (共 5 分)